MOTOMIND Motor CAN Communication Protocol (MMMCP) Specification

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Overview

- Protocol: CAN 2.0B Extended Frame
- Identifier Length: 29-bit Extended ID
- Data Length: 8 bytes
- Baud Rate: 1 Mbps

29-bit Extended ID Structure

Field	Bits	Description
MMMCP Command	28:24	Message type / command category
Pseudo-data	23:8	Usually host CAN ID, source info, status bits, or preset data depending on type
Destination Address	7:0	Target motor CAN ID or special code

8-byte Data Structure

Field	Size	Description
Data	8 bytes	Payload, interpretation depends on MMMCP Command

MMMCP Command 0 (0x00) - Get Device ID

Read the target motor CAN ID and 64-bit MCU unique identifier.

Request Frame (Host -> Motor)

ID Structure

Bit Range	Field Name	Description
28:24	MMMCP Command	`0x00`
23:16	Reserved	
15:8	Host CAN ID	`HOST_CAN_ID`
7:0	Target motor CAN ID	`TARGET_CAN_ID`

Data Structure

Byte Range	Field Name	Description
0:7	Reserved	

Response Frame (Motor -> Host)

ID Structure

Bit Range	Field Name	Description
28:24	MMMCP Command	`0x00`
23:16	Reserved	
15:8	Source motor CAN ID	`SOURCE_CAN_ID`
7:0	Special Code	`0xFE`

Data Structure

Byte Range	Field Name	Description
0:7	64-bit MCU unique identifier	big-endian `uint64`

MMMCP Command 1 (0x01) - MIT Mode Command

Dynamic motion control command in the default operation-control mode.

Request Frame (Host -> Motor)

ID Structure

Bit Range	Field Name	Description
28:24	MMMCP Command	`0x01`
23:8	Packed torque command	Big-endian `uint16`, mapped to `-17` to `17` N·m
7:0	Target motor CAN ID	`TARGET_CAN_ID`

Data Structure

Byte Range	Field Name	Description
0:1	Target angle	Big-endian `uint16`, maps to `-4*pi` to `4*pi` rad
2:3	Target angular velocity	Big-endian `uint16`, maps to `-44` to `44` rad/s
4:5	`Kp`	Big-endian `uint16`, maps to `0.0` to `500.0`
6:7	`Kd`	Big-endian `uint16`, maps to `0.0` to `5.0`

Response Frame (Motor -> Host)

- Same as MMMCP Command 2 (0x02) - Motor Feedback.

MMMCP Command 2 (0x02) - Motor Feedback

Standard feedback frame returned by several commands.

Request Frame (Host -> Motor)

- N/A

Response Frame (Motor -> Host)

ID Structure

Bit Range	Field Name	Description
28:24	MMMCP Command	`0x02`
23:22	Mode status	`0=idle`, `1=calibration`, `2=run`
21:16	Fault flags	nonzero when fault exists
15:8	Current motor CAN ID	`SOURCE_CAN_ID`
7:0	Host CAN ID	`HOST_CAN_ID`

Fault Bits in ID[21:16]

Bit	Meaning
21	Uncalibrated
20	Stall / gridlock overload fault
19	Magnetic encoder fault
18	Overtemperature
17	Three-phase overcurrent fault
16	Undervoltage fault

Data Structure

Byte Range	Field Name	Description
0:1	Current angle	Big-endian `uint16`, maps to `-4*pi` to `4*pi` rad
2:3	Current angular velocity	Big-endian `uint16`, maps to `-44` to `44` rad/s
4:5	Current torque	Big-endian `uint16`, maps to `-17` to `17` N·m
6:7	Current temperature	Big-endian `uint16`, temperature in `0.1 deg C`

MMMCP Command 3 (0x03) - Run

Transitions the motor into run state.

Request Frame (Host -> Motor)

ID Structure

Bit Range	Field Name	Description
28:24	MMMCP Command	`0x03`
23:16	Reserved	
15:8	Host CAN ID	`HOST_CAN_ID`
7:0	Target motor CAN ID	`TARGET_CAN_ID`

Data Structure

- all-zero payload.

Response Frame (Motor -> Host)

- Same as MMMCP Command 2 (0x02) - Motor Feedback.

MMMCP Command 4 (0x04) - Stop

Stops the motor. Can also clear fault when commanded accordingly.

Request Frame (Host -> Motor)

ID Structure

Bit Range	Field Name	Description
28:24	MMMCP Command	`0x04`
23:16	Reserved	
15:8	Host CAN ID	`HOST_CAN_ID`
7:0	Target motor CAN ID	`TARGET_CAN_ID`

Data Structure

- Normal stop: payload cleared to all zero
- Fault clear: `Byte0 = 1`

Response Frame (Motor -> Host)

- Same as MMMCP Command 2 (0x02) - Motor Feedback.

MMMCP Command 6 (0x06) - Set Zero

Sets the current position as mechanical zero.

Request Frame (Host -> Motor)

ID Structure

Bit Range	Field Name	Description
28:24	MMMCP Command	`0x06`
23:16	Reserved	
15:8	Host CAN ID	`HOST_CAN_ID`
7:0	Target motor CAN ID	`TARGET_CAN_ID`

Data Structure

- `Byte0 = 1`
- Other bytes zero

Response Frame (Motor -> Host)

- Same as MMMCP Command 2 (0x02) - Motor Feedback.

MMMCP Command 7 (0x07) - Set Motor CAN ID

Changes the current motor CAN ID immediately.

Request Frame (Host -> Motor)

ID Structure

Bit Range	Field Name	Description
28:24	MMMCP Command	`0x07`
23:16	New CAN ID	`NEW_CAN_ID`
15:8	Host CAN ID	`HOST_CAN_ID`
7:0	Target motor CAN ID	`TARGET_CAN_ID`

Data Structure

- N/A

Response Frame (Motor -> Host)

- Same as MMMCP Command 0 (0x00) - Get Device ID.

MMMCP Command 17 (0x11) - Single Parameter Read

Reads one entry from the parameter table.

Request Frame (Host -> Motor)

ID Structure

Bit Range	Field Name	Description
28:24	MMMCP Command	`0x11`
23:16	Reserved	
15:8	Host CAN ID	`HOST_CAN_ID`
7:0	Target motor CAN ID	`TARGET_CAN_ID`

Data Structure

Byte Range	Field Name	Description
0:1	Index	Little-endian `uint16` parameter index
2:3	Reserved	`uint16`, fixed `0x0000`
4:7	Reserved	`uint32`, fixed `0x00000000`

Response Frame (Motor -> Host)

ID Structure

Bit Range	Field Name	Description
28:24	MMMCP Command	`0x11`
23:16	Read result	`0x00` success, `0x01` failure
15:8	Motor CAN ID	`SOURCE_CAN_ID`

Bit Range	Field Name	Description
7:0	Host CAN ID	`HOST_CAN_ID`

Data Structure

Byte Range	Field Name	Description
0:1	Index	Little-endian `uint16` echoed parameter index
2:3	Reserved	`uint16`, fixed `0x0000`
4:7	Parameter data	Little-endian raw value; decode as the parameter type

MMMCP Command 18 (0x12) - Volatile Parameter Write

Writes one parameter value. This write is volatile.

Request Frame (Host -> Motor)

ID Structure

Bit Range	Field Name	Description
28:24	MMMCP Command	`0x12`
23:16	Reserved	
15:8	Host CAN ID	`HOST_CAN_ID`
7:0	Target motor CAN ID	`TARGET_CAN_ID`

Data Structure

Byte Range	Field Name	Description
0:1	Index	Little-endian `uint16` parameter index
2:3	Reserved	Little-endian `uint16`, fixed `0x0000`
4:7	Parameter data	Little-endian raw value; encode as the parameter type

Response Frame (Motor -> Host)

- Same as MMMCP Command 2 (0x02) - Motor Feedback.

MMMCP Command 21 (0x15) - Fault Feedback Frame

Reports detailed fault and warning values.

Request Frame (Host -> Motor)

- N/A

Response Frame (Motor -> Host)

ID Structure

Bit Range	Field Name	Description
28:24	MMMCP Command	`0x15`
23:16	Reserved	
15:8	Motor CAN ID	`SOURCE_CAN_ID`
7:0	Host CAN ID	`HOST_CAN_ID`

Data Structure

Byte Range	Field Name	Description
0:3	Fault value	Little-endian `uint32` bitfield, nonzero means fault
4:7	Warning value	Little-endian `uint32` bitfield

Fault Bits

Bit	Meaning
16	A-phase current sampling overcurrent
14	Stall / I2t overload protection

Bit	Meaning
9	Position initialization fault
8	Hardware identification fault
7	Encoder uncalibrated
5	C-phase current sampling overcurrent
4	B-phase current sampling overcurrent
3	Overvoltage fault
2	Undervoltage fault
1	Driver chip fault
0	Motor overtemperature fault

Warning Bits

Bit	Meaning
0	Motor overtemperature warning

MMMCP Command 22 (0x16) - Non-Volatile Parameter Write

Saves writable parameters so they persist after power cycle.

Request Frame (Host -> Motor)

ID Structure

Bit Range	Field Name	Description
28:24	MMMCP Command	`0x16`
23:16	Reserved	
15:8	Host CAN ID	`HOST_CAN_ID`
7:0	Target motor CAN ID	`TARGET_CAN_ID`

Data Structure

Byte Range	Field Name	Description
0:1	Index	Little-endian `uint16` parameter index
2:3	Reserved	Little-endian `uint16`, fixed `0x0000`
4:7	Parameter data	Little-endian raw value; encode as the parameter type

Response Frame (Motor -> Host)

- Same as MMMCP Command 2 (0x02) - Motor Feedback.

MMMCP Command 23 (0x17) - Change Motor Baudrate

- Changes the motor CAN baudrate.
- Takes effect after power cycle.

Request Frame (Host -> Motor)

ID Structure

Bit Range	Field Name	Value
28:24	MMMCP Command	`0x17`
23:16	Reserved	
15:8	Host CAN ID	`HOST_CAN_ID`
7:0	Target motor CAN ID	`TARGET_CAN_ID`

Data Structure

Byte Range	Field Name	Description
0:5	Reserved	
6	Baudrate	`F_CMD`
7	Reserved	

`F_CMD` Values

Value	Baud Rate
`0x01`	1 Mbps
`0x02`	500 kbps
`0x03`	250 kbps
`0x04`	125 kbps

Response Frame (Motor -> Host)

- Same as MMMCP Command 2 (0x02) - Motor Feedback.

MMMCP Command 24 (0x18) - Periodic Reporting Control

Enables or disables periodic reporting.

Request Frame (Host -> Motor)

ID Structure

Bit Range	Field Name	Value
28:24	MMMCP Command	`0x18`
23:16	Reserved	
15:8	Host CAN ID	`HOST_CAN_ID`
7:0	Target motor CAN ID	`TARGET_CAN_ID`

Data Structure

Byte Range	Field Name	Description
0:5	Reserved	
6	Periodic Reporting	`F_CMD`
7	Reserved	

`F_CMD` Values

Value	Meaning
`0x00`	Disable Periodic Reporting (default)
`0x01`	Enable Periodic Reporting

Notes

- Default report interval is 10 ms.
- Report interval is configured through parameter `0x7026 EPScan_time`.
- Enabled reports use MMMCP Command 24 response format documented by the manual, which mirrors the same status-bearing style used elsewhere.

Response Frame (Motor -> Host)

ID Structure

Bit Range	Field Name	Description
28:24	MMMCP Command	`0x18`
23:22	Mode status	`0=idle`, `1=calibration`, `2=run`
21:16	Fault flags	nonzero when fault exists
15:8	Current motor CAN ID	`SOURCE_CAN_ID`
7:0	Host CAN ID	`HOST_CAN_ID`

Fault Bits in ID[21:16]

Bit	Meaning
21	Uncalibrated
20	Stall / gridlock overload fault
19	Magnetic encoder fault
18	Overtemperature
17	Three-phase overcurrent fault
16	Undervoltage fault

Data Structure

Byte Range	Field Name	Value
0:1	Current angle	Big-endian `uint16`, maps to `-4*pi` to `4*pi` rad
2:3	Current angular velocity	Big-endian `uint16`, maps to `-44` to `44` rad/s
4:5	Kp	Big-endian `uint16`, maps to `0.0` to `500.0`
6:7	Kd	Big-endian `uint16`, maps to `0.0` to `5.0`

MMMCP Command 25 (0x19) - Protocol Switch (Deprecated)

Changes protocol type after reboot.

Request Frame (Host -> Motor)

ID Structure

Bit Range	Field Name	Value
28:24	MMMCP Command	`0x19`
23:16	Reserved	
15:8	Host CAN ID	`HOST_CAN_ID`
7:0	Target motor CAN ID	`TARGET_CAN_ID`

Data Structure

Byte Range	Field Name	Description
0:5	Reserved	
6	Protocol	`F_CMD`
7	Reserved	

`F_CMD` Values

Value	Meaning
`0x00`	MMMCP protocol
`0x01`	CANopen
`0x02`	MIT protocol

Response Frame (Motor -> Host)

- Same as MMMCP Command 0 (0x00) - Get Device ID.

MMMCP Command 26 (0x1A) - Version Read

Reads motor firmware version information.

Request Frame (Host -> Motor)

ID Structure

Bit Range	Field Name	Value
28:24	MMMCP Command	`0x1A`
23:16	Reserved	
15:8	Host CAN ID	`HOST_CAN_ID`
7:0	Target motor CAN ID	`TARGET_CAN_ID`

Data Structure

Byte Range	Field Name	Description
0:1	Special code	0x00, 0xC4
2:7	Reserved	-

Response Frame (Motor -> Host)

ID Structure

Bit Range	Field Name	Description
28:24	MMMCP Command	`0x1A`
15:8	Current motor CAN ID	`SOURCE_CAN_ID`
21:16	Fault flags	nonzero when fault exists
23:22	Mode status	`0=idle` , `1=calibration` , `2=run`

Bit Range	Field Name	Description
7:0	Host CAN ID	`HOST_CAN_ID`

Fault Bits in ID[21:16]

Bit	Meaning
21	Uncalibrated
20	Stall / gridlock overload fault
19	Magnetic encoder fault
18	Overtemperature
17	Three-phase overcurrent fault
16	Undervoltage fault

Data Structure

Byte Range	Field Name	Description
0:2	Special code	0x00, 0xC4, 0x56
3:6	Firmware Version	Big-endian `uint32`
7	Reserved	-

Parameter Table

for MMMCP Command 17 (0x11) / MMMCP Command 18 (0x12) / MMMCP Command 22 (0x16)

Index	Name	Description	Type	Bytes	Access
`0x7005`	`run_mode`	`0=MIT`, `1=PP`, `2=velocity`, `3=current`, `5=CSP`	`uint8`	1	R/W
`0x7006`	`iq_ref`	Current-mode Iq command	`32-bit float`	4	R/W
`0x700A`	`spd_ref`	Velocity command	`32-bit float`	4	R/W
`0x700B`	`limit_torque`	Torque limit	`32-bit float`	4	R/W
`0x7010`	`cur_kp`	Current-loop Kp	`32-bit float`	4	R/W
`0x7011`	`cur_ki`	Current-loop Ki	`32-bit float`	4	R/W
`0x7014`	`cur_filt_gain`	Current filter gain	`32-bit float`	4	R/W
`0x7016`	`loc_ref`	Position command	`32-bit float`	4	R/W
`0x7017`	`limit_spd`	CSP speed limit	`32-bit float`	4	R/W
`0x7018`	`limit_cur`	Velocity/position current limit	`32-bit float`	4	R/W
`0x7019`	`mechPos`	Mechanical position	`32-bit float`	4	R
`0x701A`	`iqf`	Filtered current	`32-bit float`	4	R
`0x701B`	`mechVel`	Mechanical velocity	`32-bit float`	4	R
`0x701C`	`VBUS`	Bus voltage	`32-bit float`	4	R
`0x701E`	`loc_kp`	Position-loop Kp	`32-bit float`	4	R/W
`0x701F`	`spd_kp`	Speed-loop Kp	`32-bit float`	4	R/W

Index	Name	Description	Type	Bytes	Access
`0x7020`	`spd_ki`	Speed-loop Ki	`32-bit float`	4	R/W
`0x7021`	`spd_filt_gain`	Speed filter gain	`32-bit float`	4	R/W
`0x7022`	`acc_rad`	Velocity-mode acceleration	`32-bit float`	4	R/W
`0x7024`	`vel_max`	PP-mode speed	`32-bit float`	4	R/W
`0x7025`	`acc_set`	PP-mode acceleration	`32-bit float`	4	R/W
`0x7026`	`EPScan_time`	Active report period selector	`uint16`	2	R/W
`0x7028`	`canTimeout`	CAN timeout threshold	`uint32`	4	R/W
`0x7029`	`zero_sta`	`0=0..2*pi`, `1=-pi..pi` power-on range	`uint8`	1	R/W
`0x702A`	`damper`	Anti-backdrive damping switch	`uint8`	1	R/W
`0x702B`	`add_offset`	Zero offset	`float`	4	R/W

Reading `loc_kp (0x701E)`:

- Request
  - MMMCP Command: `0x11`
  - Host ID: `0xFD`
  - Target motor ID: `0x7F`
  - Data: `1E 70 00 00 00 00 00 00`
- Response
  - MMMCP Command: `0x11`
  - Motor ID: `0x7F`
  - Host ID: `0xFD`

- Data: `1E 70 00 00 00 00 F0 41`

`00 00 F0 41` is IEEE-754 float `30.0` in little-endian format.

Control Mode Sequences

MIT Mode

1. Write `run_mode = 0` using MMMCP Command 18.
2. Send MMMCP Command 3 enable command.
3. Send MMMCP Command 1.
4. Receive MMMCP Command 2 feedback.

The manual describes the control law as:

```
t_ref = Kd * (v_set - v_actual) + Kp * (p_set - p_actual) + t_ff
```

Current Mode

1. Write `run_mode = 3`.
2. Send MMMCP Command 3 enable command.
3. Write `iq_ref` using MMMCP Command 18.

Velocity Mode

1. Write `run_mode = 2`.
2. Send MMMCP Command 3 enable command.
3. Write `limit_cur`.
4. Write `acc_rad`.
5. Write `spd_ref`.

Position Mode (CSP)

1. Write `run_mode = 5`.
2. Send MMMCP Command 3 enable command.
3. Write `limit_spd`.
4. Write `loc_ref`.

Position Mode (PP)

1. Write `run_mode = 1`.

2. Send MMMCP Command 3 enable command.
3. Write `vel_max`.
4. Write `acc_set`.
5. Write `loc_ref`.

Notes

Periodic Reporting

- Disabled by default.
- Enable with MMMCP Command 24.
- Interval defaults to 10 ms and is controlled by `EPScan_time`.

Zero Flag

- Controlled by `zero_sta`.
- `0`: power-on position range `0` to `2*pi`
- `1`: power-on position range `-pi` to `pi`
- Save with MMMCP Command 22 if non-volatile is required.

Power-Off Anti-Backdrive Protection

- Default behavior adds damping if the shaft is rotated quickly while unpowered.
- Disable by setting `damper = 1`.

Zero Calibration Behavior

- Supported in CSP and MIT modes.
- Not supported in PP mode.

Position Offset

- `add_offset` shifts the logical zero point by a radian offset.

End of document